Differential Equation of Simple Harmonic Oscillation & its solution

MD. GULAM MUSTAFA
Lecturer (Physics)
Department of Sc. & Hum.
Bangladesh Army International University of Science and Technology (BAIUST)
Syedpur, Cumilla-3501

Whatever we will discuss

- ✓ Definition:
  - a) Periodic Motion
  - b) Oscillatory Motion
  - c) Simple Harmonic Motion
- ✓ Differential Equation of Simple Harmonic Oscillation (SHO)
- ✓ Solution of Differential Equation of Simple Harmonic Oscillation (SHO)

Periodic Motion

Periodic motion refers to a repetitive and regular motion of an object, where it undergoes a complete cycle or oscillation over a specific time interval.

In other words, the motion repeats itself at regular intervals, and each cycle consists of the same sequence of events.

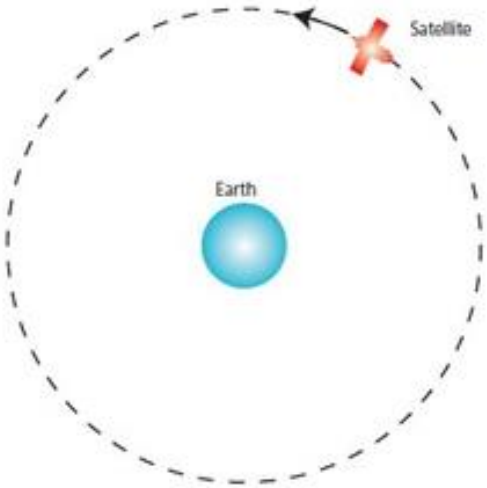
- Key characteristics of periodic motion include:

Time period (T): The time it takes for one complete cycle of the motion to occur. It is often measured in seconds.

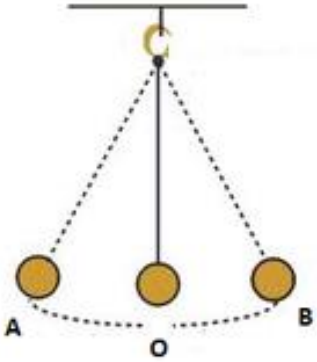
Frequency (f): The number of cycles per unit of time, typically measured in hertz (Hz), where 1 Hz equals one cycle per second. Frequency is the reciprocal of time period, meaning $f = \frac{1}{T}$.

Amplitude (a): The maximum displacement or distance from the equilibrium position that the object reaches during its oscillation.

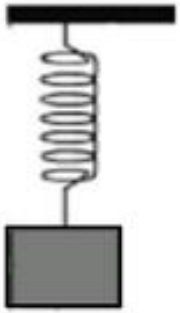
Circular motion



Circular and periodic motion of a satellite around the earth.



Periodic motion of pendulum

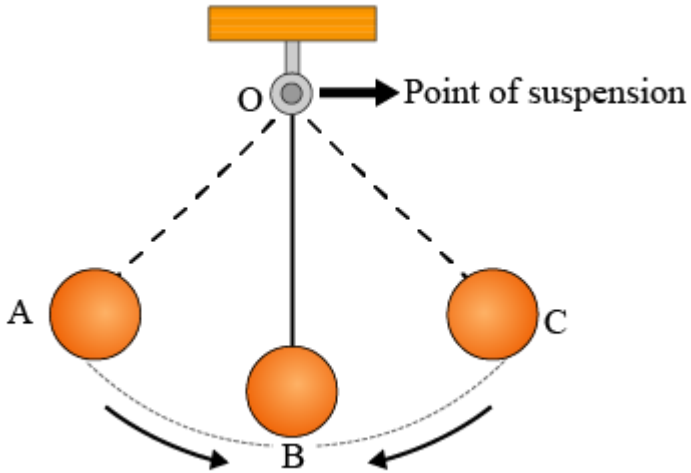


Periodic motion of spring

Oscillatory Motion

Oscillatory motion refers to repetitive back-and-forth movement around a central point or equilibrium position.

In oscillatory motion, an object or a system undergoes a regular pattern of alternating displacement from a reference point. This movement typically involves a periodic change in direction, where the object or system moves away from the equilibrium position, reaches a maximum displacement, returns to the equilibrium, and then moves in the opposite direction, reaching a maximum displacement in that direction.



Simple Harmonic Motion

Simple Harmonic Motion (SHM) is a type of periodic motion that occurs when a restoring force proportional to the displacement from an equilibrium position acts on an object. In simple terms, it is a back-and-forth movement where the restoring force is directly proportional to the displacement but opposite in direction, causing the object to oscillate about a central point.

The defining equation for simple harmonic motion is:

$$F = -ky$$

Here,

F is the restoring force,

k is the force constant (also known as the spring constant),

y is the displacement of the object from its equilibrium position.

The negative sign indicates that the restoring force acts in the opposite direction to the displacement.

❑ Key characteristics of simple harmonic motion:

- **Periodic Motion:** Simple Harmonic Motion is a type of periodic motion, meaning that it repeats itself at regular intervals.
- **Linear Restoring Force:** The motion is governed by a restoring force and the restoring force is linearly proportional to the displacement, creating a linear relationship between force and displacement.
- **Equilibrium Position:** The equilibrium position is the central point or the position where the net force is zero. In the absence of external forces, the object at equilibrium remains at rest.
- **Sinusoidal Motion:** The displacement as a function of time in simple harmonic motion follows a sinusoidal (sine or cosine) waveform.
- **Energy Interconversion:** In the absence of damping forces, the mechanical energy (kinetic energy + potential energy) of the system remains constant. The energy continually interconverts between kinetic and potential energy as the object oscillates.

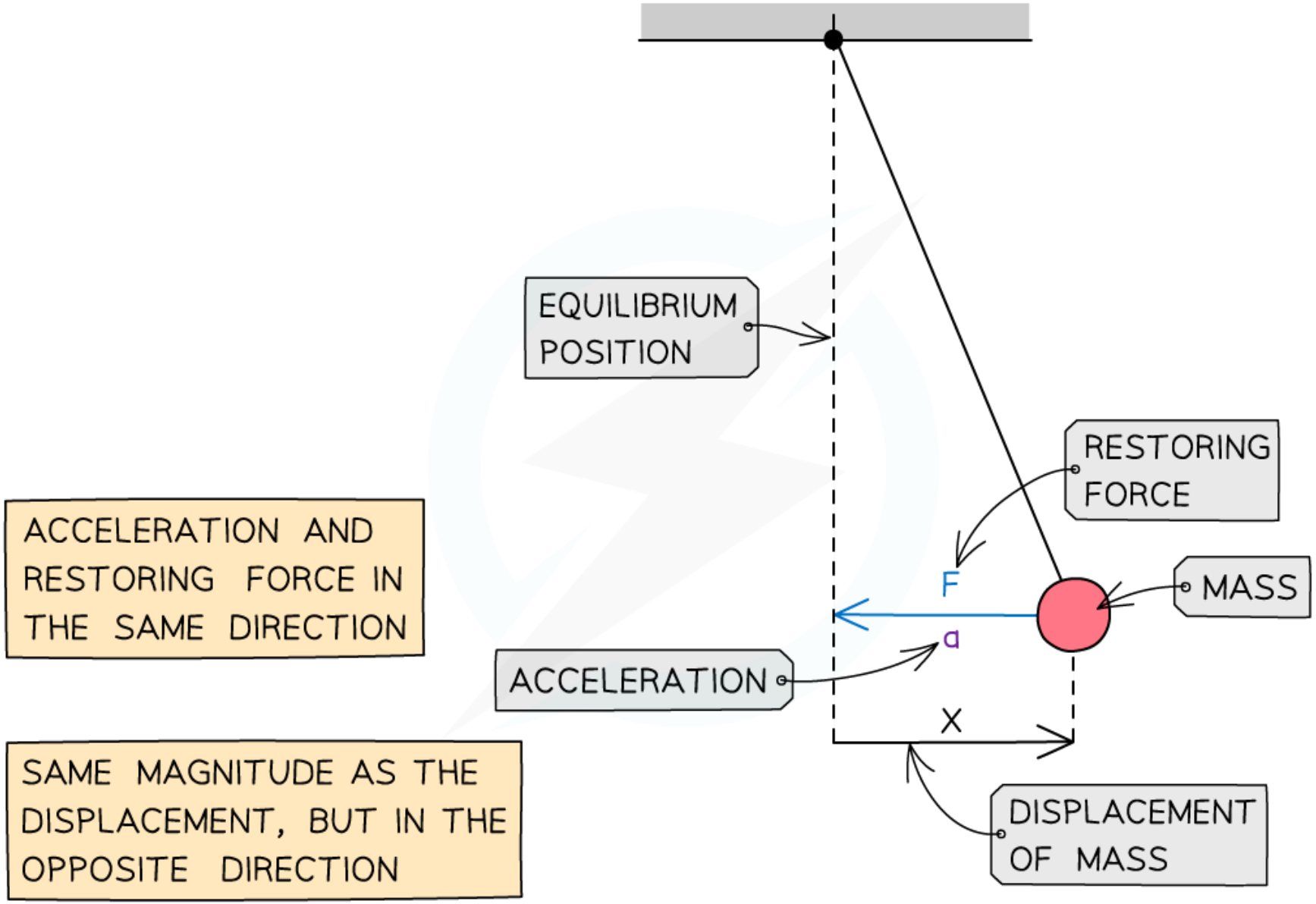


Fig: Simple harmonic motion of a simple pendulum

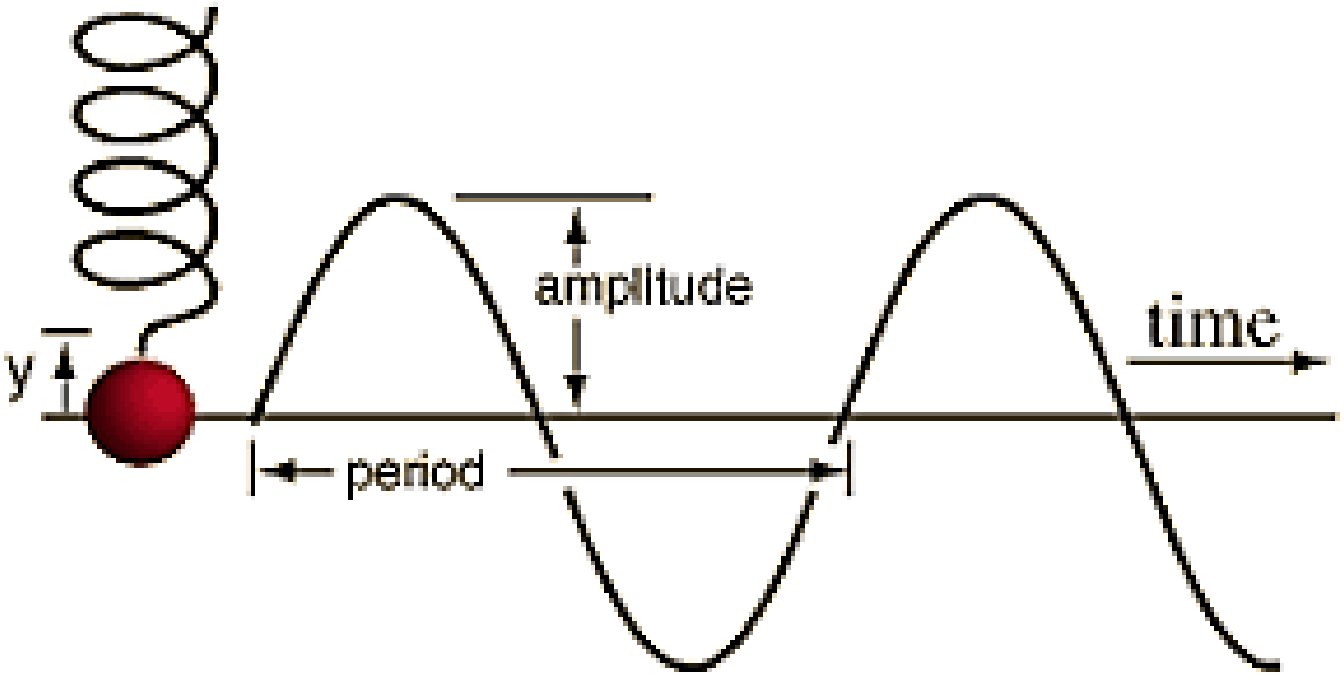


Fig: Graphical presentation of simple harmonic motion

Differential Equation of Simple Harmonic Motion

If F be the force acting on a particle and y is displacement from its mean position, then

$$F = -kx$$

according to Newton's laws of motion, $F = ma$

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

$$or, \frac{d^2x}{dt^2} = -\omega^2 x \quad \text{where}, \omega = \sqrt{\frac{k}{m}}$$

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

this is the differential equation of simple harmonic motion

Solution of the Differential Equation of SHM

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

$$\text{or, } \frac{d}{dt}\left(\frac{dx}{dt}\right) + \omega^2 x = 0$$

$$\text{or, } \frac{dv}{dt} + \omega^2 x = 0$$

$$\text{or, } \frac{dx}{dt} \cdot \frac{dv}{dx} = -\omega^2 x$$

$$\text{or, } v\,dv = -\omega^2 x\,dx$$

$$\text{or, } \int v dv = -\omega^2 \int x\,dx$$

$$\text{or, } \frac{v^2}{2} = -\omega^2 \frac{x^2}{2} + C \quad \text{..... (1)}$$

here, C = Integrating constant

If $v = 0$, then $x = A$ = amplitude

$$\therefore 0 = -\omega^2 \frac{A^2}{2} + C$$

$$\therefore C = \omega^2 \frac{A^2}{2} \quad \text{..... (2)}$$

From equation (1) and (2) we get,

$$\therefore \frac{v^2}{2} = \omega^2 \frac{x^2}{2} + \omega^2 \frac{A^2}{2}$$

$$\text{or, } v^2 = \omega^2 (A^2 - x^2)$$

$$\therefore v = \omega \sqrt{A^2 - x^2}$$

$$\text{or, } \frac{dx}{dt} = \omega \sqrt{A^2 - x^2}$$

$$\text{or, } \frac{dx}{\sqrt{A^2 - x^2}} = \omega\, dt$$

Integrating both the sides we get,

$\sin^{-1} \frac{x}{A} = \omega t + \delta$, here δ = Integration constant

$$\text{or, } \frac{x}{A} = \sin(\omega t + \delta)$$

$$\therefore x = A \sin(\omega t + \delta)$$

This equation is the solution of the differential equation of simple harmonic motion.

Thank You

Thanks for your attention